

Beam-Domain Channel Estimation for mmWave MIMO using Sub-6 GHz Out-of-Band Information

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Abstract—Future wireless multiple-input multiple-output (MIMO) systems will integrate sub-6 GHz and millimeter wave (mmWave) bands to support high data rates for latency-critical applications. The coexistence of these bands enables the use of reliable out-of-band sub-6 GHz information to assist mmWave link establishment, particularly for channel estimation. In this paper, we propose a novel beam-domain channel estimation method for mmWave MIMO that leverages sub-6 GHz beam-domain information to improve estimation accuracy. We compare the proposed method with conventional in-band methods. Simulation results show that our proposed method outperforms existing baselines in terms of spectral efficiency, in both line-of-sight (LOS) and non-line-of-sight (NLOS) scenarios.

Index Terms—millimeter wave, MIMO, digital beamforming, channel estimation, out-of-band information.

I. INTRODUCTION

Mainstream wireless systems mainly operate within the sub-6 GHz frequency range. However, these bands are increasingly unable to satisfy the growing demand for higher data rates, due to limited available spectrum. In contrast, millimeter wave (mmWave) bands (24 GHz – 300 GHz) provide substantially larger bandwidth, which enable significantly higher data rates [1]. Therefore, mmWave technology is widely considered a key enabler for next-generation communications [2], [3].

Despite their advantages, mmWave systems also introduce several challenges. Owing to their much shorter wavelengths compared to sub-6 GHz signals, mmWave transmissions experience significantly higher propagation losses [1]. To mitigate this frequency-dependent path loss and maintain a reliable link margin, mmWave systems typically employ large antenna arrays [4]. This, in turn, enables the use of multiple-input multiple-output (MIMO) techniques with highly directional beamforming and spatial multiplexing. However, exploiting large antenna arrays introduces the additional challenge of link establishment. This process, which relies on beam selection or channel estimation, becomes particularly challenging at mmWave frequencies due to the inherently low pre-beamforming signal-to-noise ratio (SNR) [4].

In addition, mmWave systems are being deployed in conjunction with sub-6 GHz systems to support multi-band operation and enhance coverage [5]. Due to their lower propagation losses, sub-6 GHz signals achieve higher pre-beamforming SNR. Furthermore, recent multi-band measurement studies have revealed that both frequency ranges exhibit comparable multipath behavior, particularly in the delay and angular

domains [6]–[9]. Hence, these cross-band similarities can be exploited to improve mmWave link establishment by utilizing reliable out-of-band information from the sub-6 GHz band.

So far, variety methods have been proposed to exploit sub-6 GHz out-of-band information to support mmWave MIMO link establishment [10]–[12]. However, these methods mainly target analog or hybrid beamforming architectures, which rely on sequential beam sweeping and incur significant time overhead, in particular when considering large antenna arrays on both link ends. In contrast, fully digital architectures enable parallel beamforming, substantially reducing time overhead and making them attractive for latency-critical applications [13]. Despite these advantages, the use of out-of-band information in fully digital mmWave systems remains underexplored. Existing work [14], [15] shows performance gains primarily under line-of-sight (LOS) conditions, with limited improvements in non-line-of-sight (NLOS) scenarios. Therefore, advanced solutions are needed to ensure reliable performance in both propagation environments.

Contribution: In this paper, we propose a novel channel estimation method for mmWave MIMO systems. The proposed method operates in the beam domain and leverages beam-domain estimates from the sub-6 GHz band to improve estimation accuracy in the mmWave band. We evaluate its performance via Monte-Carlo simulations in terms of spectral efficiency (SE), considering both LOS and NLOS propagation conditions characterized by the Rician K -factor.

Organization: The remainder of this paper is organized as follows. In Section II, we present the system model, while in Section III, we introduce the considered channel estimation methods. Section IV provides a performance evaluation based on simulations and Section V concludes the paper.

Notation: The superscript $(\cdot)^{(b)}$ indicates frequency-band-dependent quantities, where $b \in \{s, m\}$. Here, s refers to the sub-6 GHz band and m to the mmWave band. Scalars are denoted by lowercase letters x , column vectors by bold lowercase letters $\mathbf{x}$ and matrices by bold uppercase letters $\mathbf{X}$. The i -th row and j -th column of $\mathbf{X}$ are represented by $\mathbf{X}_{i,:}$ and $\mathbf{X}_{:,j}$, respectively. The Kronecker product is denoted by $\otimes$ and matrix vectorization by $\text{vec}(\cdot)$. The superscripts, $(\cdot)^T$, $(\cdot)^*$ and $(\cdot)^H$ correspond to transpose, complex conjugation and Hermitian transpose, respectively. The Frobenius norm is written as $\|\cdot\|_F$, while $\|\cdot\|$ denotes the Euclidean norm. Finally, $\delta[\cdot]$ represents the discrete delta function.

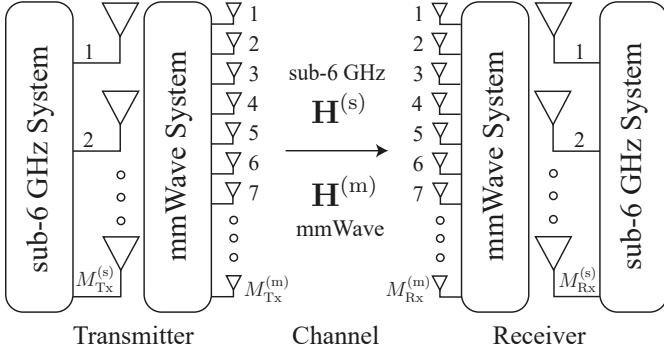


Fig. 1. The considered multi-band MIMO system employs co-located antenna arrays that operate in both sub-6 GHz and mmWave frequency bands.

II. SYSTEM MODEL

We consider a point-to-point multi-band MIMO system where sub-6 GHz and mmWave subsystems operate simultaneously under far-field conditions. The transmitter is equipped with $M_{Tx}^{(s)}$ sub-6 GHz and $M_{Tx}^{(m)}$ mmWave antenna elements. Similarly, the receiver employs $M_{Rx}^{(s)}$ sub-6 GHz and $M_{Rx}^{(m)}$ mmWave antennas, as illustrated in Fig. 1. The sub-6 GHz and mmWave arrays are co-located, aligned uniform linear arrays (ULAs)¹ of dipole elements with half-wavelength spacing, leading to identical LOS angle-of-departure (AoD) ϑ_{LOS} and angle-of-arrival (AoA) φ_{LOS} across the two bands. We assume perfect synchronization in both time and frequency, among all transmit and receive antennas. Furthermore, each antenna is connected to a dedicated radio frequency (RF) chain, allowing fully digital beamforming in both bands. The considered multi-band system operates in time-division duplex (TDD) mode, ensuring reciprocal channel responses.

A. Multi-Band Channel Model

In this work, we consider channel estimation in the mmWave band by leveraging out-of-band information obtained from the sub-6 GHz band. Therefore, it is essential to follow an appropriate simulation methodology capable of generating frequency-dependent channels across multiple bands. In this work, we adopt the 3rd Generation Partnership Project (3GPP) channel model [17] to ensure consistent and realistic multi-frequency simulations. More specifically, the considered 3GPP channel model enables the joint generation of channels at different carrier frequencies while preserving the inherent spatial correlation of propagation paths.

For a given delay tap τ , the channel is represented by a complex-valued matrix $\mathbf{H}^{(b)}[\tau] \in \mathbb{C}^{M_{Rx}^{(b)} \times M_{Tx}^{(b)}}$, based on the equivalent baseband formulation of an orthogonal frequency-division multiplexing (OFDM) system. We consider

a frequency-selective, time-invariant channel in the delay domain, modeled according to a Rician fading process as [18]

$$\mathbf{H}^{(b)}[\tau] = \sqrt{\eta^{(b)}} \sqrt{\frac{\kappa^{(b)}}{1 + \kappa^{(b)}}} \underbrace{\mathbf{H}_{fs}^{(b)} \delta[0]}_{\text{free-space}} + \sqrt{\eta^{(b)}} \sqrt{\frac{1}{1 + \kappa^{(b)}}} \underbrace{\mathbf{H}_{sp}^{(b)}[\tau]}_{\text{stochastic part}}. \quad (1)$$

In (1), the parameter $\eta^{(b)}$ represents large-scale effects such as path loss and shadow fading, as described in detail in [14]. The parameter $\kappa^{(b)}$ denotes the Rician K -factor for the corresponding frequency band. To capture the distinct propagation behavior across frequencies, the mmWave K -factor is modeled as a scaled version of its sub-6 GHz counterpart, i.e., $\kappa^{(m)} = c_\kappa \kappa^{(s)}$, where c_κ denotes the scaling factor. Finally, the channel representation in the frequency domain for subcarrier n , denoted by $\mathbf{H}^{(b)}[n]$, is obtained by applying the discrete Fourier transform (DFT) to the delay-domain channel $\mathbf{H}^{(b)}[\tau]$.

The deterministic LOS component is modeled according to a free-space far-field propagation assumption and is represented by

$$\mathbf{H}_{fs}^{(b)} = e^{j\chi^{(b)}} \mathbf{a}_{Rx}^{(b)}(\varphi_{LOS}) \left(\mathbf{a}_{Tx}^{(b)}(\vartheta_{LOS}) \right)^H, \quad (2)$$

where

$$\mathbf{a}_{Tx}^{(b)}(\vartheta_{LOS}) = \begin{bmatrix} 1 & \dots & e^{-j2\pi(M_{Tx}^{(b)}-1)\frac{\Delta d^{(b)}}{\lambda^{(b)}}\sin(\vartheta_{LOS})} \end{bmatrix}^T \quad (3)$$

and

$$\mathbf{a}_{Rx}^{(b)}(\varphi_{LOS}) = \begin{bmatrix} 1 & \dots & e^{-j2\pi(M_{Rx}^{(b)}-1)\frac{\Delta d^{(b)}}{\lambda^{(b)}}\sin(\varphi_{LOS})} \end{bmatrix}^T \quad (4)$$

denote the transmit and receiver array steering vectors, which correspond to the AoD ϑ_{LOS} and AoA φ_{LOS} , respectively. Additionally, a random phase term $\chi^{(b)} \sim \mathcal{U}(-\pi, \pi)$ is included in (2) to account for the random phase offset at the first (reference) antenna element.

The stochastic part of the channel is given by the channel matrix $\mathbf{H}_{sp}^{(b)}[\tau]$, which is obtained by sampling its continuous-time counterpart, defined as

$$\mathbf{H}_{sp}^{(b)}(\tau) = \sum_{c=1}^C \sum_{r_c=1}^{R_c^{(b)}} \alpha_{r_c}^{(b)} \delta\left(\tau - \tau_c - \tau_{r_c}^{(b)}\right) \times \mathbf{a}_{Rx}^{(b)}\left(\varphi_c + \varphi_{r_c}^{(b)}\right) \left(\mathbf{a}_{Tx}^{(b)}\left(\vartheta_c + \vartheta_{r_c}^{(b)}\right) \right)^H. \quad (5)$$

In (5), C represents the number of clusters. Each cluster is characterized by a mean delay τ_c , a mean AoA φ_c and a mean AoD ϑ_c . Within each cluster, multiple propagation paths (or rays) are present, with $R_c^{(b)}$ rays connecting the transmitter and receiver. Each individual ray r_c is associated with a complex path gain $\alpha_{r_c}^{(b)}$, along with small perturbations around the cluster means, including a relative delay shift $\tau_{r_c}^{(b)}$, a relative AoA shift $\varphi_{r_c}^{(b)}$ and a relative AoD shift $\vartheta_{r_c}^{(b)}$. The

¹The analysis in [16] shows that co-located and aligned sub-6 GHz and mmWave ULAs are practically feasible, with only minimal mutual impact on their radiation characteristics.

generation of these parameters in (5) follows the 3GPP channel modeling framework, which adapts them according to the selected propagation scenario.

Furthermore, for multi-frequency channel generation described in [17], the 3GPP channel model ensures partial consistency across frequency bands. In particular, large-scale cluster characteristics (i.e., number of clusters C , mean delays τ_c , mean angular parameters φ_c and ϑ_c) are shared between the sub-6 GHz and mmWave frequency bands. In contrast, small-scale properties, including cluster powers described as complex path gains $\alpha_{r_c}^{(b)}$, as well as the intra-cluster variations (i.e., relative delay shifts $\tau_{r_c}^{(b)}$, relative AoA shifts $\varphi_{r_c}^{(b)}$ and relative AoD shifts $\vartheta_{r_c}^{(b)}$) are frequency-dependent and are therefore generated independently for each frequency band. Further details on the multi-band channel generation procedure can be found in the 3GPP channel model specification described in [17].

B. Link Establishment

Establishing a reliable communication link involves two main phases: a training phase followed by a data transmission phase. During the training phase, the wireless channel is estimated at both sub-6 GHz and mmWave frequency bands. These estimates are then utilized jointly to enable efficient data transmission over the mmWave link.

1) *Training Phase:* In the training phase, pilot symbols are transmitted isotropically in space over both frequency bands to acquire initial channel estimates. Each transmit antenna is assigned a set of pilot symbols that are known at the receiver. The pilot symbols $\phi^{(b)}[n] \in \mathbb{C}^{M_{Tx}^{(b)} \times 1}$, drawn from a quadrature amplitude modulation (QAM) constellation, are distributed across $N^{(b)}$ subcarriers such that different antennas occupy mutually exclusive (non-overlapping) subsets of subcarriers. For the t -th transmit antenna, where $t \in \{1, \dots, M_{Tx}^{(b)}\}$, the pilot allocation is given by

$$\phi_t^{(b)}[n] = \begin{cases} \phi^{(b)}[n], & n \in \{t, t + M_{Tx}^{(b)}, \dots, N^{(b)} - M_{Tx}^{(b)} + t\} \\ 0, & \text{else} \end{cases} \quad (6)$$

The input-output relationship of the training phase is given by

$$\mathbf{y}^{(b)}[n] = \mathbf{H}^{(b)}[n]\phi^{(b)}[n] + \mathbf{w}^{(b)}[n], \quad (7)$$

where $\mathbf{y}^{(b)}[n] \in \mathbb{C}^{M_{Rx}^{(b)} \times 1}$ represents the received signal and $\mathbf{w}^{(b)}[n] \sim \mathcal{CN}(0, \sigma_{w^{(b)}}^2 \mathbf{I}_{M_{Rx}^{(b)}})$ denotes the additive white Gaussian noise (AWGN) with the power of $\sigma_{w^{(b)}}^2$. Based on the received pilot symbols, least-squares (LS) estimates of the channel are first obtained at the pilot subcarrier locations. To reconstruct the channel over the entire bandwidth, linear interpolation is applied across the remaining subcarriers, yielding an estimated channel matrix $\hat{\mathbf{H}}^{(b)}[n] \in \mathbb{C}^{M_{Rx}^{(b)} \times M_{Tx}^{(b)}}$. As the channel estimate is initially obtained at the receiver, it must also be made available at the transmitter to enable optimal beamforming design. Since the system operates using the TDD protocol, channel acquisition at the transmitter is achieved by exploiting channel reciprocity in both frequency bands.

2) *Data Transmission:* In the data transmission phase, communication is carried out exclusively over the mmWave band. The channel estimates obtained during the training phase are further processed using the methods, which will be introduced in Section III. The resulting mmWave channel estimate $\bar{\mathbf{H}}^{(m)}[n]$ is then used to design the precoder $\bar{\mathbf{F}}^{(m)}[n]$ and combiner $\bar{\mathbf{Q}}^{(m)}[n]$. To maximize the SE of the MIMO system, we employ singular value decomposition (SVD) of the channel matrix. The compact-form SVD of $\bar{\mathbf{H}}^{(m)}[n]$ is given by

$$\bar{\mathbf{H}}^{(m)}[n] = \bar{\mathbf{Q}}^{(m)}[n] \bar{\Sigma}^{(m)}[n] \left(\bar{\mathbf{F}}^{(m)}[n] \right)^H, \quad (8)$$

where $\bar{\mathbf{Q}}^{(m)}[n] \in \mathbb{C}^{M_{Rx}^{(m)} \times \ell_{\max}}$ and $\bar{\mathbf{F}}^{(m)}[n] \in \mathbb{C}^{M_{Tx}^{(m)} \times \ell_{\max}}$ are semi-unitary matrices containing the left and right singular vectors, respectively. The diagonal matrix $\bar{\Sigma}^{(m)}[n]$ consists of singular values, with each diagonal entry $\bar{\sigma}_{(i)}^{(m)}[n]$ representing the i -th largest singular value of the channel. Assuming a full-rank channel, the maximum number of spatial data streams is $\ell_{\max} = \min(M_{Rx}^{(m)}, M_{Tx}^{(m)})$. Power allocation across the parallel data streams is controlled by a diagonal matrix $\bar{\mathbf{P}}^{(m)}[n]$. To achieve the maximum achievable rate, the transmit power is distributed according to the water-filling principle [18]. The total transmit power constraint

$$\left\| \bar{\mathbf{F}}^{(m)}[n] \left(\bar{\mathbf{P}}^{(m)}[n] \right)^{1/2} \right\|_F^2 = P_T^{(m)} \quad (9)$$

is satisfied by the precoder for all subcarriers.

Let $\mathbf{x}^{(m)}[n] \in \mathbb{C}^{\ell_{\max} \times 1}$ denote the transmitted symbol vector. The input-output relationship for the data transmission phase can be expressed as

$$\begin{aligned} \mathbf{y}^{(m)}[n] &= \left(\bar{\mathbf{Q}}^{(m)}[n] \right)^H \mathbf{H}^{(m)}[n] \bar{\mathbf{F}}^{(m)}[n] \left(\bar{\mathbf{P}}^{(m)}[n] \right)^{1/2} \mathbf{x}^{(m)}[n] \\ &\quad + \left(\bar{\mathbf{Q}}^{(m)}[n] \right)^H \mathbf{w}^{(m)}[n], \end{aligned} \quad (10)$$

where $\mathbf{y}^{(m)}[n] \in \mathbb{C}^{\ell_{\max} \times 1}$ is the received symbol vector. Owing to the semi-unitary property of $\bar{\mathbf{Q}}^{(m)}[n]$, the transformed noise $\left(\bar{\mathbf{Q}}^{(m)}[n] \right)^H \mathbf{w}^{(m)}[n]$ follows the same statistical distribution as $\mathbf{w}^{(m)}[n]$.

III. CHANNEL ESTIMATION METHODS

In this section, we first revisit conventional in-band channel estimation methods for the mmWave system. In particular, we describe the LS estimator introduced in Section II-B1, followed by the linear minimum mean square error (LMMSE) estimator. Subsequently, we then introduce the proposed out-of-band aided beam-domain channel estimation method.

A. Least-Squares (LS)

The LS estimator serves as a low-complexity baseline for mmWave channel estimation. It relies solely on in-band observations and does not exploit any prior statistical information

about the channel. Specifically, an initial mmWave estimate $\tilde{\mathbf{H}}^{(m)}[n]$ is obtained at pilot subcarriers using LS estimation, followed by linear interpolation across frequency to obtain the channel on all subcarriers, as detailed in Section II-B1. The resulting channel estimate $\bar{\mathbf{H}}^{(m)}[n]$ is given by

$$\bar{\mathbf{H}}^{(m)}[n] = \tilde{\mathbf{H}}^{(m)}[n]. \quad (11)$$

B. Linear Minimum Mean Squared Error (LMMSE)

To improve estimation accuracy, we consider the LMMSE approach which incorporates second-order channel statistics. This method uses the LS estimate $\tilde{\mathbf{H}}^{(m)}[n]$ as an initial input and refines it by incorporating the spatial covariance matrix of the channel and the noise variance. The resulting estimate is given by

$$\bar{\mathbf{H}}^{(m)}[n] = \tilde{\mathbf{R}}_{\mathbf{H}}^{(m)} \left(\tilde{\mathbf{R}}_{\mathbf{H}}^{(m)} + \sigma_{w^{(m)}}^2 \mathbf{I}_{M_{\text{Rx}}^{(m)}} \right)^{-1} \tilde{\mathbf{H}}^{(m)}[n]. \quad (12)$$

In (12), $\tilde{\mathbf{R}}_{\mathbf{H}}^{(m)}$ represents the estimated spatial covariance matrix, which is given by

$$\tilde{\mathbf{R}}_{\mathbf{H}}^{(m)} = \frac{1}{N^{(m)} M_{\text{Tx}}^{(m)}} \sum_{n=1}^{N^{(m)}} \tilde{\mathbf{H}}^{(m)}[n] \left(\tilde{\mathbf{H}}^{(m)}[n] \right)^{\text{H}}. \quad (13)$$

In contrast to the LS estimator, the LMMSE estimator requires additional statistical information, namely the channel covariance and noise variance, and incurs higher computational complexity.

C. Out-of-Band Aided Beam-Domain Estimation (OBABE)

This method leverages channel estimates from both the mmWave and sub-6 GHz bands to improve channel estimation performance at mmWave. In particular, the sub-6 GHz out-of-band channel estimate is exploited to identify the dominant angular components of the propagation channel, which are then employed as prior information to filter out low-power and noise-dominated components in the mmWave beam-domain estimate. The overall procedure is described as follows.

We first transform the frequency-domain channel estimates $\tilde{\mathbf{H}}^{(b)}$, for both-frequency bands b , into the beam domain. Assuming ULAs, the beam-domain representation is given by

$$\tilde{\mathbf{H}}_{\text{B}}^{(b)}[n] = \frac{1}{\sqrt{M_{\text{Tx}}^{(b)} M_{\text{Rx}}^{(b)}}} \left(\mathbf{U}_{\text{Rx}}^{(b)} \right)^{\text{H}} \tilde{\mathbf{H}}^{(b)}[n] \mathbf{U}_{\text{Tx}}^{(b)}, \quad (14)$$

where the matrices $\mathbf{U}_{\text{Tx}}^{(b)} \in \mathbb{C}^{M_{\text{Tx}}^{(b)} \times M_{\text{Tx}}^{(b)}}$ and $\mathbf{U}_{\text{Rx}}^{(b)} \in \mathbb{C}^{M_{\text{Rx}}^{(b)} \times M_{\text{Rx}}^{(b)}}$ are DFT matrices, respectively. These DFT matrices consist of orthogonal steering vectors that span the angular domain, enabling a spatial decomposition of the channel into distinct beam directions.

Next, we average the sub-6 GHz beam-domain estimates across subcarriers, which is given by

$$\hat{\mathbf{H}}_{\text{B}}^{(s)} = \frac{1}{N^{(s)}} \sum_{n=1}^{N^{(s)}} \tilde{\mathbf{H}}_{\text{B}}^{(s)}[n], \quad (15)$$

where $\hat{\mathbf{H}}_{\text{B}}^{(s)} \in \mathbb{C}^{M_{\text{Rx}}^{(s)} \times M_{\text{Tx}}^{(s)}}$ denotes the averaged beam-domain estimate. This averaging reduces frequency-dependent fluctuations and yields a stable representation of the dominant spatial structure. However, since the antenna dimensions at mmWave are typically equal to or larger than those at sub-6 GHz, the resulting reference must be adapted accordingly. To this end, the averaged sub-6 GHz beam-domain estimate is upsampled to match the mmWave antenna configuration, resulting in

$$\hat{\mathbf{H}}_{\text{B}}^{(m)} = \hat{\mathbf{H}}_{\text{B}}^{(s)} \otimes \mathbf{1}_{c_{\text{Rx}} \times c_{\text{Tx}}}. \quad (16)$$

The upsampling factors $c_{\text{Tx}} = M_{\text{Tx}}^{(m)} / M_{\text{Tx}}^{(s)}$ and $c_{\text{Rx}} = M_{\text{Rx}}^{(m)} / M_{\text{Rx}}^{(s)}$ account for the difference in the number of transmit and receive antennas between the mmWave and sub-6 GHz systems, respectively. This ensures that both representations are defined over the same beam-space grid, enabling direct comparison and filtering. In practical deployments, antenna array sizes are typically chosen as powers of two. Accordingly, we assume that the sub-6 GHz antenna configuration is an integer divisor of the corresponding mmWave configuration. Next, we vectorize the upsampled reference beam-domain matrix, given by

$$\hat{\mathbf{h}}_{\text{B}}^{(m)} = \text{vec} \left(\hat{\mathbf{H}}_{\text{B}}^{(m)} \right) \in \mathbb{C}^{M^{(m)} \times 1}, \quad (17)$$

where $M^{(m)} = M_{\text{Tx}}^{(m)} M_{\text{Rx}}^{(m)}$ represents the total number of mmWave beam-domain coefficients. Further, we calculate the power contribution of each beam and normalize it as $\hat{\mathbf{p}}_{\text{B}}^{(m)} = \left| \hat{\mathbf{h}}_{\text{B}}^{(m)} \right|^2 / \left\| \hat{\mathbf{h}}_{\text{B}}^{(m)} \right\|^2$. To identify dominant beam components, we sort $\hat{\mathbf{p}}_{\text{B}}^{(m)}$ in descending order and then compute the corresponding cumulative power distribution, yielding

$$\hat{\mathbf{c}}_{\text{B}}^{(m)} = \mathbf{L}^{(m)} \mathbf{\Pi}^{(m)} \hat{\mathbf{p}}_{\text{B}}^{(m)}, \quad (18)$$

where $\mathbf{\Pi}^{(m)} \in \mathbb{R}^{M^{(m)} \times M^{(m)}}$ represents the permutation matrix which sorts the elements of $\hat{\mathbf{p}}_{\text{B}}^{(m)}$ in descending order and

$$\mathbf{L}^{(m)} = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 1 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & \cdots & 1 \end{bmatrix} \in \mathbb{R}^{M^{(m)} \times M^{(m)}} \quad (19)$$

denotes the lower triangular matrix that performs the cumulative summation. Based on this, we determine the minimum number of dominant beams $\hat{i}_{\text{B}}$, given by

$$\hat{i}_{\text{B}} = \min \left\{ i \in \{1, \dots, M^{(m)}\} : \hat{c}_{\text{B},i}^{(m)} > p \right\}, \quad (20)$$

whose cumulative power accounts for a fraction $p \in [0, 1]$ of the total power. A higher value of p retains more mmWave beams, while a lower value results in stronger beam sparsification. We then extract the corresponding index set $\hat{\mathcal{I}}_{\text{B}}$ of the $\hat{i}_{\text{B}}$ dominant beams from $\hat{\mathbf{p}}_{\text{B}}^{(m)}$, given by

$$\hat{\mathcal{I}}_{\text{B}} = \arg \text{top}_{\hat{i}_{\text{B}}} \left(\hat{\mathbf{p}}_{\text{B}}^{(m)} \right). \quad (21)$$

TABLE I
SIMULATION PARAMETERS

Parameter	Value	
Frequency Band	sub-6 GHz	mmWave
Carrier Frequency f_c	2.55 GHz	25.5 GHz
Wavelength λ	11.76 cm	1.176 cm
Bandwidth B	20.16 MHz	403.2 MHz
Subcarrier Spacing Δf	60 kHz	240 kHz
Antenna Configuration ($M_{\text{Rx}} \times M_{\text{Tx}}$)	8×8	16×16
Transmit Power P_T	30 dBm	30 dBm
Noise Figure F	3 dB	6 dB
Fraction p	N/A	0.95
Number of Realizations L_r	5000	5000

This index set is used to filter the mmWave beam-domain channel estimate. Specifically, we vectorize the instantaneous mmWave beam-domain channel for each subcarrier by

$$\tilde{\mathbf{h}}_B^{(m)}[n] = \text{vec} \left(\tilde{\mathbf{H}}_B^{(m)}[n] \right) \quad (22)$$

and retain only entries indexed by $\hat{\mathcal{L}}_B$, while setting all other components to zero, as follows:

$$\tilde{\mathbf{h}}_{B,i}^{(m)}[n] = \begin{cases} \tilde{\mathbf{h}}_{B,i}^{(m)}[n], & i \in \hat{\mathcal{L}}_B \\ 0, & \text{else} \end{cases} \quad (23)$$

Finally, we reshape the vectorized estimate back into the matrix form by

$$\bar{\mathbf{H}}_B^{(m)}[n] = \text{unvec} \left(\tilde{\mathbf{h}}_B^{(m)}[n] \right) \quad (24)$$

and subsequently transform the beam-domain channel estimate back to the frequency domain, yielding the resulting channel estimate

$$\bar{\mathbf{H}}^{(m)}[n] = \frac{1}{\sqrt{M_{\text{Tx}}^{(m)} M_{\text{Rx}}^{(m)}}} \mathbf{U}_{\text{Rx}}^{(m)} \bar{\mathbf{H}}_B^{(m)}[n] \left(\mathbf{U}_{\text{Tx}}^{(m)} \right)^H. \quad (25)$$

IV. SIMULATION-BASED COMPARISON

To assess the performance of the proposed channel estimation methods, we simulate the achievable SE over a frequency-selective channel. The key simulation parameters are listed in Tab. I. For modeling the frequency-selective channel, we adopt the 3GPP model for the indoor-office scenario with an adjustable K -factor, following the methodology described in [17]. Due to the strong similarity between K -factors across bands [19], we set the scaling parameter to $c_\kappa = 1$ (i.e., 0 dB), such that the mmWave K -factor satisfies $\kappa^{(m)} = \kappa^{(s)}$. As a reference, we first analyze conventional LS and LMMSE channel estimation methods that rely solely on mmWave observations, as introduced in Section III-A and Section III-B. In addition, we include the performance with perfect channel state information (CSI), i.e., $\bar{\mathbf{H}}^{(m)}[n] = \mathbf{H}^{(m)}[n]$.

We consider the achievable SE expressed in bits/s/Hz as the main performance metric. Since this work focuses on the impact of channel estimation quality on achievable data

rates and all considered methods employ identical training overhead, the training overhead is not included in the SE calculation. Throughout the analysis, we assume equal power allocation across all subcarriers. The achievable SE averaged over $N^{(m)}$ subcarriers is given by

$$\text{SE} = \frac{1}{N^{(m)}} \sum_{n=1}^{N^{(m)}} \sum_{\mu=1}^{\ell_{\max}} \log_2 (1 + \text{SINR}_\mu[n]), \quad (26)$$

where the effective signal-to-interference-and-noise ratio (SINR) for the stream μ is denoted by [20]

$$\text{SINR}_\mu[n] = \frac{\left| \bar{\mathbf{G}}_{\mu,\mu}^{(m)}[n] \right|^2}{\sum_{\substack{\nu=1 \\ \nu \neq \mu}}^{\ell_{\max}} \left| \bar{\mathbf{G}}_{\mu,\nu}^{(m)}[n] \right|^2 + \left(\sigma_\mu^2 + \sigma_{w^{(m)}}^2 \right) \left\| \bar{\mathbf{Q}}_{:, \mu}^{(m)}[n] \right\|^2}. \quad (27)$$

The elements $\bar{\mathbf{G}}_{\mu,\nu}^{(m)}[n]$, with $\mu, \nu \in \{1, \dots, \ell_{\max}\}$, denote the entries of the channel gain matrix $\bar{\mathbf{G}}^{(m)}[n] \in \mathbb{C}^{\ell_{\max} \times \ell_{\max}}$ for the n -th subcarrier, which is given by

$$\bar{\mathbf{G}}^{(m)}[n] = \left(\bar{\mathbf{Q}}^{(m)}[n] \right)^H \mathbf{H}^{(m)}[n] \bar{\mathbf{F}}^{(m)}[n] \left(\bar{\mathbf{P}}^{(m)}[n] \right)^{1/2}. \quad (28)$$

Moreover, σ_μ^2 corresponds to the diagonal elements (variance) of the estimation error covariance matrix $\bar{\mathbf{C}}_\epsilon^{(m)} \in \mathbb{C}^{\ell_{\max} \times \ell_{\max}}$ averaged over subcarrier, given by

$$\bar{\mathbf{C}}_\epsilon^{(m)} = \frac{1}{N^{(m)} \ell_{\max}} \sum_{n=1}^{N^{(m)}} \bar{\epsilon}^{(m)}[n] \left(\bar{\epsilon}^{(m)}[n] \right)^H. \quad (29)$$

The estimation error matrix is given by $\bar{\epsilon}^{(m)}[n] = \mathbf{G}^{(m)}[n] - \bar{\mathbf{G}}^{(m)}[n]$. Finally, $\bar{\epsilon}^{(m)}[n]$ represents the mismatch between the channel gain matrix obtained using the estimated precoder/combiner and the one obtained under perfect CSI.

We evaluate the performance of the proposed channel estimation methods as a function of the Rician K -factor. The simulation results are presented in terms of achievable SE in Fig. 2.

As expected, channel estimation based on LS generally yields inferior SE performance compared to LMMSE-based estimation due to the absence of statistical prior information. However, when incorporating out-of-band information via the proposed out-of-band aided beam-domain estimation (OBABE) method, a consistent SE improvement is observed over conventional in-band estimation, irrespective of the K -factor or the baseline estimator. For the LS-based implementation, OBABE provides substantial gains over its in-band counterpart. In particular, the SE improves by approximately 25% in the low K -factor regime (−20 dB, NLOS conditions) and up to 53% in the high K -factor regime (20 dB, LOS conditions). In contrast, when OBABE is applied on top of LMMSE estimation, the relative gains are more moderate, as the LMMSE baseline already achieves strong performance. Specifically, the SE gain is around 11% for a low K -factor of −20 dB and increases to approximately 15% for a high K -factor of 20 dB.

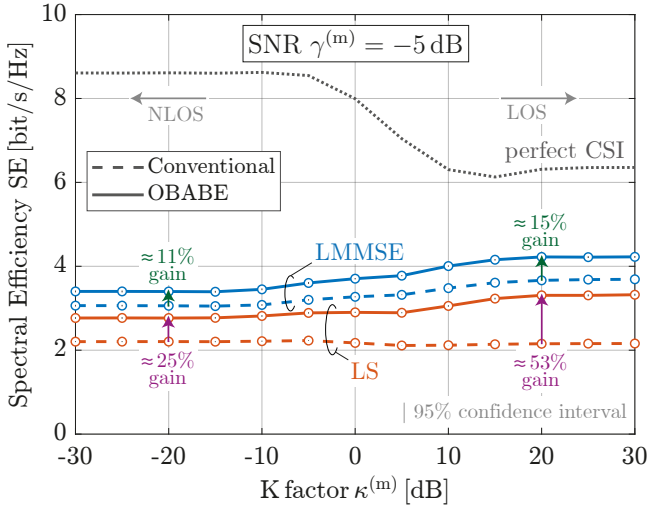


Fig. 2. Incorporating out-of-band information via the proposed OBABE method brings consistent improvements over conventional in-band estimation, irrespective of the K -factor or the baseline estimator.

Furthermore, we observe higher gains in the LOS regime (i.e., at higher K -factor). The primary reason for this behavior is the increased spatial sparsity under LOS conditions, where the channel energy is predominantly concentrated in a small number of beam components, typically dominated by the direct path. In this case, the number of selected dominant beams is approximately $\hat{i}_B = 35$, corresponding to only 14% of the total beams. In contrast, in the NLOS regime (lower K -factors), the channel exhibits reduced sparsity, with energy distributed across multiple channel paths. Consequently, the number of selected dominant beams increases to approximately $\hat{i}_B = 113$, corresponding to 44% of the total beams. Moreover, the dominant propagation paths are less consistent across frequencies, which limits the predictive capability of sub-6 GHz information for mmWave.

V. CONCLUSION

In this paper, we propose a novel method for beam-domain CSI estimation in mmWave MIMO systems by exploiting out-of-band beam-domain CSI obtained from the sub-6 GHz band. Numerical results demonstrate the effectiveness of the proposed method compared to conventional approaches that rely solely on in-band mmWave information, achieving consistent performance gains in both NLOS and LOS channel conditions.

REFERENCES

- [1] T. S. Rappaport, S. Sun, R. Mayzus, H. Zhao, Y. Azar, K. Wang, G. N. Wong, J. K. Schulz, M. Samimi, and F. Gutierrez, "Millimeter wave mobile communications for 5G cellular: It will work!" *IEEE Access*, vol. 1, pp. 335–349, 2013.
- [2] A. F. Molisch, C. F. Mecklenbräuker, T. Zemen, A. Prokes, M. Hofer, F. Pasic, and H. Hammoud, "Millimeter-wave V2X channel measurements in urban environments," *IEEE Open Journal of Vehicular Technology*, vol. 6, pp. 520–541, 2025.
- [3] H. Hammoud, Y. Zhang, Z. Cheng, S. Sangodoyin, M. Hofer, F. Pasic, T. M. Pohl, R. Závorka, A. Prokes, T. Zemen, C. F. Mecklenbräuker, and A. F. Molisch, "Double-directional V2V channel measurement using ReRoMA at 60 GHz," *IEEE Transactions on Vehicular Technology*, pp. 1–14, 2025.
- [4] R. W. Heath, N. González-Prelcic, S. Rangan, W. Roh, and A. M. Sayeed, "An overview of signal processing techniques for millimeter wave MIMO systems," *IEEE Journal of Selected Topics in Signal Processing*, vol. 10, no. 3, pp. 436–453, 2016.
- [5] M. Hofer, F. Pasic, B. Rainer, J. Blumenstein, A. Prokes, C. F. Mecklenbräuker, A. F. Molisch, and T. Zemen, "Enabling vehicular mmWave communication links using cmWave information," in *2025 19th European Conference on Antennas and Propagation (EuCAP)*, 2025.
- [6] S. Sangodoyin, U. T. Virk, D. Burghal, K. Haneda, and A. F. Molisch, "Joint characterization of mm-wave and cm-wave device-to-device fading channels," in *MILCOM 2018 - 2018 IEEE Military Communications Conference (MILCOM)*, 2018, pp. 194–200.
- [7] M. Hofer, D. Löschbrand, F. Pasic, D. Radovic, B. Rainer, J. Blumenstein, C. F. Mecklenbräuker, S. Sangodoyin, H. Hammoud, G. Matz, A. F. Molisch, and T. Zemen, "Similarity of wireless multiband propagation in urban vehicular-to-infrastructure scenarios," in *2024 IEEE 35th International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC)*, 2024.
- [8] H. Miao, J. Zhang, P. Tang, L. Tian, X. Zhao, B. Guo, and G. Liu, "Sub-6 GHz to mmWave for 5G-advanced and beyond: Channel measurements, characteristics and impact on system performance," *IEEE Journal on Selected Areas in Communications*, vol. 41, no. 6, pp. 1945–1960, 2023.
- [9] F. Pasic, M. Mussbah, M. Hofer, S. Caban, S. Schwarz, T. Zemen, M. Rupp, and C. F. Mecklenbräuker, "From sub-6 GHz to millimeter-wave: Dynamic indoor measurements, channel characteristics and performance evaluation," *IEEE Open Journal of the Communications Society*, vol. 6, pp. 8190–8208, 2025.
- [10] T. Nitsche, A. B. Flores, E. W. Knightly, and J. Widmer, "Steering with eyes closed: Mm-wave beam steering without in-band measurement," in *2015 IEEE Conference on Computer Communications (INFOCOM)*, 2015, pp. 2416–2424.
- [11] A. Ali, N. González-Prelcic, and R. W. Heath, "Spatial covariance estimation for millimeter wave hybrid systems using out-of-band information," *IEEE Transactions on Wireless Communications*, vol. 18, no. 12, pp. 5471–5485, 2019.
- [12] M. S. Sim, Y.-G. Lim, S. H. Park, L. Dai, and C.-B. Chae, "Deep learning-based mmWave beam selection for 5G NR/6G with sub-6 GHz channel information: Algorithms and prototype validation," *IEEE Access*, vol. 8, pp. 51 634–51 646, 2020.
- [13] C. Liu, M. Li, L. Zhao, P. Whiting, S. V. Hanly, and I. B. Collings, "Millimeter-wave beam search with iterative deactivation and beam shifting," *IEEE Transactions on Wireless Communications*, vol. 19, no. 8, pp. 5117–5131, 2020.
- [14] F. Pasic, M. Hofer, M. Mussbah, S. Sangodoyin, S. Caban, S. Schwarz, T. Zemen, M. Rupp, A. F. Molisch, and C. F. Mecklenbräuker, "Millimeter wave MIMO channel estimation using sub-6 GHz out-of-band information," *IEEE Transactions on Communications*, vol. 73, no. 11, pp. 10 692–10 708, 2025.
- [15] F. Pasic, L. Eller, S. Schwarz, M. Rupp, and C. F. Mecklenbräuker, "Deep learning-based mmWave MIMO channel estimation using sub-6 GHz channel information: CNN and UNet approaches," in *IEEE INFOCOM 2025 - IEEE Conference on Computer Communications Workshops (INFOCOM WKSHPS)*, 2025.
- [16] F. Pasic, J. Soklič, R. Langwieser, S. Schwarz, and C. F. Mecklenbräuker, "Multi-band patch antenna array for out-of-band aided millimeter wave communication," in *2026 IEEE International Workshop on Antenna Technology (iWAT)*, 2026.
- [17] 3GPP, "Study on channel model for frequencies from 0.5 to 100 GHz," 3rd Generation Partnership Project (3GPP), Technical report (TR) 38.901, 2022, version 17.0.0.
- [18] A. F. Molisch, *Wireless communications: From Fundamentals to Beyond 5G*, 3rd ed. IEEE Press - Wiley, 2023.
- [19] F. Pasic, M. Hofer, T. Zemen, A. F. Molisch, and C. F. Mecklenbräuker, "Time-varying Rician K-factor in measured vehicular channels at cmWave and mmWave bands," in *2025 19th European Conference on Antennas and Propagation (EuCAP)*, 2025.
- [20] A. Kammoun, A. Müller, E. Björnson, and M. Debbah, "Linear precoding based on polynomial expansion: Large-scale multi-cell MIMO systems," *IEEE Journal of Selected Topics in Signal Processing*, vol. 8, no. 5, pp. 861–875, 2014.